

Reasons for the growth in the data storage solutions segment

Let's check out some of the reasons for this:

1. Various recent government regulations, like Sarbanes-Oxley, ask businesses to maintain and keep a backup of different types of data which they might have otherwise deleted.
2. Many of the small businesses have now started archiving different e-mail messages, even those dating back five or more years for various legal reasons.
3. Also, the pervasiveness of spyware and viruses requires backups and that again requires more storage capacity.
4. There has been a growing need to back up and store different large media files, such as video, MP3, etc, and make the same available to users on a specific network. This is again generating a demand for large storage solutions.
5. Each newer version of any software application or operating system demands more space and memory than its predecessor, which is another reason driving the demand for large storage solutions.

Different types of storage options

There are different types of storage solutions that can be used based on individual requirements, as listed below.

Flash memory thumb drives: These drives are particularly useful to mobile professionals since they consume little power, are small enough to even fit on a keychain and have almost no moving parts. You can connect any Flash memory thumb drive to your laptop's Universal Serial Bus (USB) port and back up different files on the system. Some of the USB thumb drives also provide encryption to protect files in case the drive gets lost or is stolen. Flash memory thumb drives also let us store our Outlook data (like recent e-mails or calendar items), different bookmarks on Internet Explorer, and even some of the desktop applications. That way, you can leave your laptop at home and just plug the USB drive into any borrowed computer to access all your data elsewhere.

External hard drives: An inexpensive and relatively simpler way to add more memory storage is to connect an external hard drive to your computer. External hard disk drives that are directly connected to PCs have several disadvantages. Any file stored only on the drive but not elsewhere requires to be backed up. Also, if you travel somewhere for work and need access to some of the files on an external drive, you will have to take the drive with you or remember to make a copy of the required files to your laptop's internal drive, a USB thumb drive, a CD or any other storage media. Finally, in case of a fire or other catastrophe at your place of business, your data will not be completely protected if it's stored on an external hard drive.

Online storage: There are different services which provide remote storage and backup over the Internet. All such services offer businesses a number of benefits. By

backing up your most important files to a highly secure remote server, you are actually protecting the data stored at your place of business. You can also easily share different large files with your clients, partners or others by providing them with password-protected access to your online storage service, hence eliminating the need to send those large files by e-mail. And in most cases, you can log into your account from any system using a Web based browser, which is one of the great ways to retrieve files when you are away from your PC. Remote storage can be a bit slow, especially during an initial backup session, and only as fast as the speed of your network's access to that storage. For extremely large files, you may require higher speed network access.

Network attached storage: Network attached storage (NAS) provides fast, reliable and simple access to data in any IP networking environment. Such solutions are quite suitable for small or mid-sized businesses that require large volumes of economical storage which can be shared by multiple users over a network. Given that many of the small businesses lack IT departments, this storage solution is easy to deploy, can be managed and consolidated centrally. This type of storage solution can be as simple as a single hard drive with an Ethernet port or even built-in Wi-Fi connectivity.

More sophisticated NAS solutions can also provide additional USB as well as FireWire ports, enabling you to connect external hard drives to scale up the overall storage capacity of businesses. A NAS storage solution can also offer print-server capabilities, which let multiple users easily share a single printer. A NAS solution may also include multiple hard drives in a Redundant Array of Independent Disks (RAID) Level 1 array. This storage system contains two or more equivalent hard drives (similar to two 250GB drives) in a single network-connected device. Files written to the first (main) drive are automatically written to the second drive as well. This kind of automated redundancy present in NAS solutions implies that if the first hard drive dies, we will still have access to all our applications and files present on the second drive. Such solutions can also help in offloading files being served by other servers on your network, which increases the performance. A NAS system allows you to consolidate storage, hence increasing the efficiency and reducing costs. It simplifies the storage administration, data backup and its recovery, and also allows for easy scaling to meet the growing storage needs.

Choosing the right storage solution

There are a number of storage solutions available in the market, which meet diverse requirements. At times, you could get confused while trying to choose the right one. Let's get rid of that confusion by considering some of the important aspects of a storage solution.

Scalability: This is one of the important factors to be considered while looking for any storage solution. In different distributed storage systems, storage capacity can be added in two ways. The first way involves adding disks